2024 Ph H2 Q3

Section: Our Dynamic Universe

Topic: Collisions, Explosions and Impulse

Summary of Question

A Newton’s Cradle consists of a line of suspended balls.

When one ball is lifted and released, it strikes the next ball, transmitting a force through the stationary balls, causing the ball at the opposite end to swing out.

Task: Comment on the teacher’s statement that the cradle demonstrates a number of physics principles. (3 marks)

Answer

The Newton’s Cradle demonstrates several key principles:

• Conservation of momentum: In collisions, total momentum before = total momentum after. The momentum of the incoming ball is transferred through the line of balls to the ball at the far end.

• Conservation of kinetic energy (elastic collisions): The collisions are nearly elastic, so kinetic energy is transferred efficiently to the outgoing ball.

• Newton’s third law: When balls collide, the forces they exert on each other are equal and opposite.

• Energy transformations: gravitational potential energy (when a ball is raised) ↔ kinetic energy (as it swings) ↔ sound/heat losses.

Final answer: The Newton’s Cradle illustrates momentum and energy conservation, and Newton’s third law.

Revision Tips

When asked to ‘comment’, always mention at least two relevant principles.

Newton’s Cradle is the classic demonstration of conservation of momentum and energy.

Remember: real cradles gradually stop due to non-elastic losses (sound, heat, air resistance).